

MODEL QUESTION PAPER
MASS TRANSFER OPERATIONS II

Max.Marks : 75

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. List *any two* applications of distillation.
2. Define relative volatility.
3. State Raoult's law.
4. Define the term weeping in plate column.
5. Define reflux ratio.
6. List *any two* advantages of super critical fluid extraction.
7. Define leaching.
8. Identify the significance of value of selectivity in extraction.
9. Define distribution coefficient.

Part B

II. Answer any EIGHT questions from the following. Each question carries 3 marks

8 X 3 = 24 Marks

1. List examples of maximum and minimum boiling azeotropes.
2. Describe the principle of distillation.
3. State Rayleigh's equation and mention the terms in it.
4. Identify the limitations of McCabe Thiele method.
5. Describe total reflux condition in distillation.
6. Enumerate the feed conditions and corresponding values of 'q' in fractionation.
7. Explain the factors affecting rate of solid –liquid extraction.
8. List the applications of solid-liquid extraction.
9. Draw a neat sketch of rotating disk contactor and label its parts.
10. Draw the block diagram for liquid-liquid extraction.

Part C

Answer ALL questions. Each question carries 7 marks

6 X 7 = 42 Marks

- III. Differentiate between ideal and non-ideal solutions.

OR

- IV. The vapor pressure data of n-heptane (A) and n-octane (B) are given in the following table. For a total pressure of 1 atm, assuming the mixture to behave ideally, construct the equilibrium diagram.

Temperature (K)	371.4	378	383	388	393	398.6
P_A^O (kPa)	101.3	125.3	140	160	180	205.3
P_B^O (kPa)	44.4	55.6	64.5	74.8	86.6	101.3

- V. Describe the functions of reboiler and condenser in a distillation column.

OR

- VI. Mixture containing 70 mole % toluene and 30 mole % benzene is distilled in a continuous column at ordinary pressure to give product containing 95 mole % benzene and the bottom containing 5 mole % benzene. Reflux ratio is 3. Feed is saturated liquid. Calculate the number of theoretical plate required and determine the feed plate.

x	0.06	0.16	0.26	0.38	0.51	0.66	0.82	1
y	0.13	0.31	0.45	0.6	0.72	0.83	0.92	1

- VII. Describe the constructional details of sieve tray.

OR

- VIII. Liquid mixture containing 40 mole% methanol and 60 mole% water is fed to differential distillation unit at atmospheric pressure with 60 mole% of the liquid to be distilled. Find the composition of the distillate and the residue.

Equilibrium data:

x	0.05	0.1	0.2	0.3	0.4	0.5
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y	0.27	0.42	0.57	0.66	0.73	0.78
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IX. Illustrate the working of continuous countercurrent decantation system.

OR

X. Explain heap leaching.

XI. Illustrate the working of Bollman extractor.

OR

XII. Illustrate the working of Rotocel extractor.

XIII. List and explain the important factors to be considered for selecting a suitable solvent in liquid-liquid extraction.

OR

XIV. Describe the working of spray towers used for liquid-liquid extraction.